

Weekly Diary – Week 14

Internship Organization: Vikaspedia

Duration: 22/04/2026 – 26/04/2026

Objective of the Week

The primary objective of this week was to initiate the development of the SEO Keyword Generation System by planning the architecture, configuring the development environment, and implementing the foundational backend setup. The focus was on preparing the system for efficient keyword extraction, caching, and SEO-focused processing.

Date – 22 April

On this day, the project objectives and workflow were analyzed in detail. Research was conducted on SEO keyword generation techniques, search-demand validation, and long-tail keyword optimization. The overall architecture for frontend, backend, database, and Redis caching was planned.

Date – 23 April

The backend development environment was configured using Java 21, Spring Boot, Maven, MySQL, and Redis. Required dependencies such as Spring Web, Spring Data JPA, and Redis integration libraries were added to the project. Initial project folders and package structures were also organized.

Date – 24 April

Database schema planning and implementation were completed for storing generated keywords and search analytics. MySQL tables such as keywords and search_analytics were created, and database connectivity was configured using environment variables within the Spring Boot application.

Date – 25 April

Redis caching setup was implemented to improve system performance and reduce repeated keyword processing. Cache storage strategies and key structures were planned to efficiently manage generated keyword data and trend validation results.

Date – 26 April

Basic text preprocessing functionality was implemented for keyword extraction. This included tokenization, stop-word filtering, phrase cleaning, and preparation of content data for further keyword analysis and scoring.